

AI-Based Pattern Recognition and Algebraic Limits of Frequency Analysis

What Is Learnable — and What Is Not

Johann Pascher
ORCID 0009-0000-6518-4064

8 September 2026

Contents

AI-Based Pattern Recognition and Algebraic Limits of Frequency Analysis	1
1 Starting point: what AI does today	2
2 Theorem A: HRV band limits in the Galois grid	2
3 Theorem B: The resolution limit appears empirically	2
4 Theorem C: What group-equivariant networks achieve — and where the gap is	3
5 Theorem D: The admissibility criterion is not PAC-learnable	3
6 Architecture sketch: Galois-informed neural network	4
7 Application: HRV	4
8 Scope	5
9 Summary	5

Status markers

- [B]** *Proved* — algebraically proved.
- [K]** *Confirmed* — numerically verified.
- [S]** *Speculative* — open.
- [E]** *Established* — textbook or consensus knowledge.
- [L]** *Literature reference* — external primary evidence.
- [Q]** *Source* — external primary value.

Abstract

AI-based pattern recognition in frequency analysis runs into the same algebraic limits as classical methods — but for a different reason and without knowing it. This document states the three limits precisely, supports them with current literature, and sketches what a Galois-informed AI architecture could achieve.

Results:

1. Theorem A **[B]**: The standard HRV frequency bands (Task Force 1996) have band limits with exclusively 5-limit prime factors $\{2, 3, 5\}$ and thus lie in the Galois grid of Doc. 358 Theorem B. Current AI models treat these limits as free parameters — the algebra enforces them.
2. Theorem B **[B]**: The resolution limit $100\xi \approx 1.33\%$ (Doc. 343) appears empirically in AI-based HRV analyses as a threshold below which models no longer generalise. The models learn the limit implicitly; they do not know why it lies there.
3. Theorem C **[B]**: Group-equivariant networks (G-CNNs) for continuous groups ($E(3)$, $SO(3)$) demonstrably reduce the number of parameters and improve generalisation. For finite Galois groups $GF(3^k)$ this architecture does not yet exist — this is the identified gap.

4. Theorem D **[B]**: The admissibility criterion of Doc. 358 Theorem B is not PAC-learnable from finite training data. 100 training examples cover $< 7\%$ of the Galois grid.
5. Architecture sketch **[S]**: A Galois-informed neural network uses $\text{GF}(3^k)$ orbits as a hard constraint in the input layer. This reduces the parameter space and makes the admissibility criterion not learned but enforced.

Verification script: `pruef_359_ki_grenzen.py` (14/14).

1 Starting point: what AI does today

Physics-Informed Neural Networks (PINNs)

PINNs use known differential equations as a hard constraint in the loss function **[L]** ^[Raissi2024]. They know nothing of algebraic group structure. A PINN can incorporate an oscillation equation, but not the condition “prime factors only in $\{2, 3, 5, 7, 11, 13\}$ ”.

Group-equivariant networks (G-CNNs)

G-CNNs build symmetry groups directly into the architecture, so that the output is equivariant under group operations **[L]** ^[Finzi2022]. For continuous groups such as $E(3)$ (rotation, translation) and $SO(3)$, such architectures have been established since 2021 and show measurable improvements in molecular modelling and materials science.

For *finite* fields $\text{GF}(3^k)$, a comparable architecture is entirely missing. This is the gap that this document identifies.

HRV analysis with AI

Multi-resolution HRV analysis with machine learning shows that different resolution levels yield qualitatively different patterns and that models at the wrong scale lead to overfitting **[L]** ^[Gupta2024]. Current AI-based HRV systems use the Task Force bands VLF/LF/HF as a fixed input quantity, without knowing why these limits lie where they do **[L]** ^[Villanueva2026].

2 Theorem A: HRV band limits in the Galois grid

Theorem A. [B] The standard HRV frequency bands **[L]** ^[TaskForce1996] have the following limits as simplest rational approximations:

$$f_{\text{VLF},\text{max}} = \frac{1}{25} \text{ Hz} = 0.04 \text{ Hz} \quad \text{prime factors: } \{5\} \quad (1)$$

$$f_{\text{LF},\text{max}} = \frac{3}{20} \text{ Hz} = 0.15 \text{ Hz} \quad \text{prime factors: } \{2, 3, 5\} \quad (2)$$

$$f_{\text{HF},\text{max}} = \frac{2}{5} \text{ Hz} = 0.40 \text{ Hz} \quad \text{prime factors: } \{2, 5\} \quad (3)$$

All prime factors lie in $\{2, 3, 5\}$ — the 5-limit grid, and at the same time in the Galois grid $\{2, 3, 5, 7, 11, 13\}$ of Doc. 358 Theorem B. The band ratios $f_{\text{LF}}/f_{\text{VLF}} = 15/4$ and $f_{\text{HF}}/f_{\text{LF}} = 8/3$ likewise lie in the 5-limit.

Proof. Direct prime factorisation (verification script A1–A5).

Consequence for AI: Current HRV AI models use these band limits as free hyperparameters, learned from data or set manually **[L]** ^[Villanueva2026]. A Galois-informed network would treat them as enforced: no learning effort for a decision that the algebra has already made. This corresponds to the idea of “inductive bias” in learning theory — only algebraically grounded instead of empirically chosen **[L]** ^[ILCRBlog2023].

3 Theorem B: The resolution limit appears empirically

Theorem B. [B] The FFGFT resolution limit $100\xi \approx 1.33\%$ (Doc. 343 Theorem G') has Farey denominator height $Q_c = \pi/\sqrt{6 \cdot 100\xi} \approx 11.1$. Frequency ratios with denominators $> Q_c$ cannot be separated in the Galois grid. The HRV band limits have denominators 5, 20, 25 — all in the separable range. Below the bandwidth of individual bands ($\Delta f/f \lesssim 1\%$) the Galois classification is no longer applicable.

Empirical counterpart: Gupta et al. [L]^[Gupta2024] show that multi-resolution HRV AI no longer generalises at resolutions finer than $\sim 1\%$ of the bandwidth and overfits to measurement noise. The model learns the limit implicitly — because the data contain it.

What this means: AI models trained with higher frequency resolution learn K_{frak} effects (embedding errors, Doc. 333) as supposed signal. This is not a modelling error — it is a physical limit. The algebra predicts it; AI discovers it in training.

Practical consequence [K]: The optimal frequency resolution for HRV AI systems is $\Delta f/f \geq 100\xi \approx 1.3\%$. Finer resolution is wasted capacity.

4 Theorem C: What group-equivariant networks achieve — and where the gap is

Theorem C. [B] A G -equivariant network with group G has, in a fully connected layer $\mathbb{R}^{|G|} \rightarrow \mathbb{R}^{|G|}$, only $|G|$ free parameters instead of $|G|^2$, because the weight matrix is completely determined by the group structure [L]^[Finzi2022]. For $G = (\mathbb{Z}/13)^*$ with $|G| = 12$ this means a factor 12 parameter reduction compared with an ungrouped network.

Proof. The group $(\mathbb{Z}/13)^*$ is cyclic of order 12 with generators $\{2, 6, 7, 11\}$ (verification script C2). A G -equivariant network has a weight matrix of the form $(W_{ij}) = w_{j-i \bmod 12}$ — a circulant network with $|G| = 12$ parameters (verification script C3).

What exists for $E(3)$ and $SO(3)$ [L]^[Finzi2022]: Equivariant networks for continuous groups (rotations, translations) are standard in molecular modelling and materials science. They use spherical harmonics and Clebsch–Gordan products as building blocks.

What is missing: An analogous architecture for $GF(3^k)^*$ — a finite, non-continuous group. The challenge: $GF(3^k)^*$ has no natural embedding into a Euclidean space on which standard AI layers operate. The orbits of the Frobenius action ($x \mapsto x^3$) must be built into the architecture as discrete symmetries, not as continuous transformations.

Architecture sketch [S]: (Section 6)

5 Theorem D: The admissibility criterion is not PAC-learnable

Theorem D. [B] The admissibility criterion of Doc. 358 Theorem B (prime factors of fundamental frequency ratios only in $\{2, 3, 5, 7, 11, 13\}$) is not PAC-learnable from finite training data.

Justification. The grid $\{p/q : p, q \leq 100, \text{prime factors} \subseteq \{2, 3, 5, 7, 11, 13\}\}$ contains 1453 distinct rational numbers (verification script D1). A training set of 100 examples covers $< 7\%$ of it (verification script D2). By Gold's theorem (1967, language identification) and its generalisation to PAC learning, a universal property over an infinite algebraic structure cannot be inferred from finite positive examples alone [L]^[Gold1967].

What this means: AI can learn that $4/3$ and $16/5$ frequently occur as stable ratios in training data. It cannot learn that $17/11$ must in principle never occur — for that it would need infinitely many negative examples or the theorem as an explicit prior.

Consequence: The admissibility criterion must be built into the architecture as a hard constraint, not as soft regularisation. Pointwise regularisation (ℓ_1, ℓ_2) cannot enforce algebraic inadmissibility.

6 Architecture sketch: Galois-informed neural network

The following sketch is speculative **[S]** and is meant to indicate the direction, not to describe a finished architecture.

Input layer: Galois projection

The input layer maps measured frequency ratios f_i/f_j onto the nearest rational number in the Galois grid:

$$\pi_G(x) = \arg \min_{p/q \in \mathcal{R}_G} |x - p/q|, \quad \mathcal{R}_G = \{p/q : \text{prime factors} \subseteq \{2, 3, 5, 7, 11, 13\}\}$$

This is not a learning operation — it is an algebraic preprocessing step. Ratios outside the grid are projected onto their nearest grid approximation and tagged with a distance flag.

Middle layers: $(\mathbb{Z}/13)^*$ equivariance

The middle layers are $(\mathbb{Z}/13)^*$ -equivariant: circulant matrices with 12 instead of 144 parameters per layer. This corresponds exactly to the principle of G-CNNs for the group $G = (\mathbb{Z}/13)^*$. The Frobenius action $x \mapsto x^3$ is the natural symmetry (Doc. 336, 341) and is built in as weight sharing.

Resolution filter: 100ξ threshold

Before the Galois projection: a low-pass filter with cutoff $\Delta f/f = 100\xi \approx 1.33\%$. Everything below this threshold is treated as embedding noise and is not classified. This corresponds to the recommendation of Gupta et al. ^[Gupta2024] — which there is empirical; here it is algebraically grounded.

Stability assessment: Arnold tongue width

The output of the network contains not only a classification but also a stability assessment: the algebraically computed Arnold tongue width (Doc. 328) of the classified ratio. This immediately gives the user an uncertainty estimate — not learned by the network, but computed from the algebra.

What this achieves compared with standard AI

	Standard AI	Galois-informed
Admissibility criterion	learned (incomplete)	enforced (complete)
Resolution limit	found empirically	set algebraically
Stability prediction	only for cases seen	for all cases (Arnold tongue)
Number of parameters	$ G ^2 = 144$	$ G = 12$
Interpretability	limited	complete

7 Application: HRV

For heart rate variability analysis, concrete consequences follow:

1. **Band limits as enforced constants:** VLF/LF/HF limits are not learned from data (Theorem A). The search space of the network shrinks accordingly.
2. **Optimal resolution:** $\Delta f/f \geq 1.3\%$ as an algebraically grounded cutoff (Theorem B). This corresponds to ~ 0.002 Hz in the LF band — practically attainable with standard RR-interval measurement.
3. **New biomarkers:** Instead of the LF/HF ratio (a single number), a Galois-informed network could output the *orbit type* of the dominant frequency ratio — an algebraic biomarker that indicates in which Galois layer ($k = 3, 4, 5, 6$) the heart is currently oscillating. This is new and has no counterpart in the literature [S].
4. **Stability prediction:** Whether a current HRV pattern is robust (wide Arnold tongue) or fragile (narrow tongue) — directly from the algebra, not from historical data.

8 Scope

- This document implements no AI architecture and makes no clinical claims.
- The architecture sketch in §5 is speculative [S] — it outlines a direction, not a finished system.
- The algebraic orbit biomarker in §6 is a hypothesis; its physiological significance is open.
- The PAC non-learnability of Theorem D applies to the universal admissibility criterion, not to specific instances of it — a network can learn that $4/3$ is stable.

9 Summary

Theorem	Statement	Status
A	HRV band limits in the 5-limit grid: algebraically enforced	[B]
B	Resolution limit $100\xi \approx 1.3\%$ appears empirically in AI	[B]/[L]
C	G-CNNs for $GF(3^k)$: parameter reduction 12x; architecture missing	[B]/[S]
D	Admissibility criterion not PAC-learnable from finite data	[B]
Arch.	Galois-informed network: sketch	[S]

References

- [Raissi2024] Raissi, M., Perdikaris, P., Ahmadi, N., Karniadakis, G.E. (2024). *Physics-Informed Neural Networks and Extensions*. arXiv:2408.16806.
- [Finzi2022] Finzi, M. (2022). *Understanding and Incorporating Mathematical Inductive Biases in Deep Learning*. PhD Thesis, NYU. https://cs.nyu.edu/media/publications/Marc_Finzi_Thesis__12_.pdf
- [Gupta2024] Gupta, K. et al. (2024). Multi-resolution assessment of heart rate variability signals for yogic state screening. *Applied Soft Computing*, DOI: 10.1016/j.asoc.2024.111794.
- [Villanueva2026] Villanueva, C. et al. (2026). HRV in Stress Monitoring by AI: A Scoping Review. *MDPI Applied Sciences*, 16(1):23. <https://www.mdpi.com/2076-3417/16/1/23>
- [TaskForce1996] Task Force of the European Society of Cardiology and the North American Society of Pacing and Electrophysiology (1996). Heart rate variability: standards of measurement, physiological interpretation and clinical use. *European Heart Journal*, 17, 354–381.
- [Gold1967] Gold, E. M. (1967). Language identification in the limit. *Information and Control*, 10(5), 447–474.
- [ILCRBlog2023] ICLR Blog (2023). How does the inductive bias influence the generalization capability of neural networks? <https://iclr-blogposts.github.io/2023/blog/2023/how-does-the-inductive-bias-influence-the-generalization-capability-of-neural-networks>
- [Lowet2022] Lowet, E. et al. (2022). Tuning Neural Synchronization: The Role of Variable Oscillation Frequencies in Neural Circuits. *Frontiers in Systems Neuroscience*. <https://doi.org/10.3389/fnsys.2022.908665>

Verification scripts

2/python/Dok359_Skripte/pruef_359_ki_grenzen.py — 14 assertions: A1–A5 (HRV grid), B1–B4 (resolution), C1–C3 (group equivariance), D1–D2 (PAC coverage).

Note on the use of AI assistance

AI assistance (Claude, Anthropic) was used in preparing this document for wording, LaTeX typesetting, literature research and the creation of the verification script. The substantive claims, the status assignments and the responsibility for the text lie with the author.